

S.No. : 553

NEC 4201

No. of Printed Pages : 07

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43402

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B. Tech. Examination, 2024-25

(Even Semester)

BASIC ELECTRONICS ENGINEERING

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION-A

1. Attempt all parts of the following : $8 \times 1 = 8$

- Why is Si diode preferred over Ge diode?
- Define cut-in voltage of a p-n junction.
- What is the ripple factor for full-wave rectifier?
- What is clipping circuits?
- What is meant by biasing of a transistor?

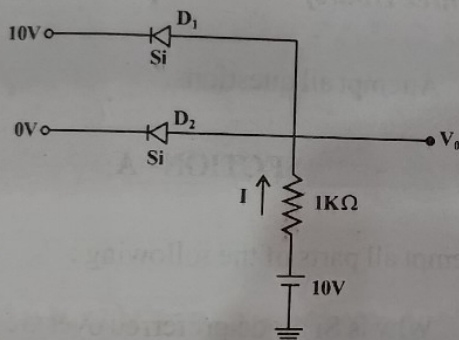
[P. T. O.

- (f) What is the major difference between a bipolar and a unipolar device?
- (g) What do you mean by term slew rate?
- (h) Explain MINTERM and MAXTERM in Boolean expression.

SECTION - B

2. Attempt any two parts of the following : $2 \times 6 = 12$

(a) Determine I & V_0 for the circuits shown in fig.



(b) A full wave centre tapped rectifier with 220V, 50 Hz sinusoidal input and turn ratio of 5 : 1 has a load resistance of 500Ω . Diode forward resistance is 20Ω . Determine :

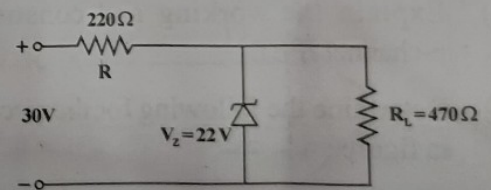
- (i) Mean or average load current

- (ii) DC output voltage
- (iii) Rectification efficiency
- (iv) Ripple factor
- (c) Explain the working and construction of n-p-n transistor.
- (d) Derive expressions for voltage gain of inverting and non-inverting ideal operational amplifier configuration.

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Determine V_L , I_L , I_Z and I_R for the circuit shown in figure :

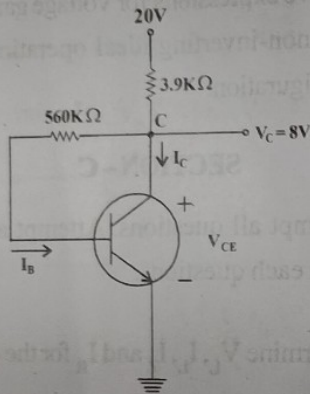


- (b) Define I_{CBO} and I_{CEO} . How are they different? Derive the relation between them.

[P.T.O.]

(c) Given $V_C = 8V$ for the network of fig, determine :

- (i) I_B
- (ii) I_C
- (iii) β
- (iv) V_{CE}



4. (a) Explain the working and construction of p-channel JFET.

(b) Determine the following for the circuit shown in figure :

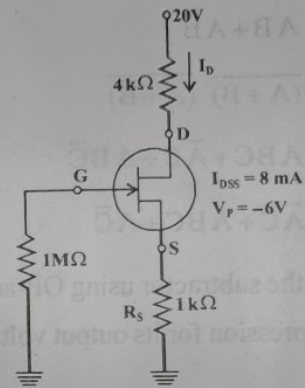
- (i) V_{GS}
- (ii) I_D

(iii) V_{DS}

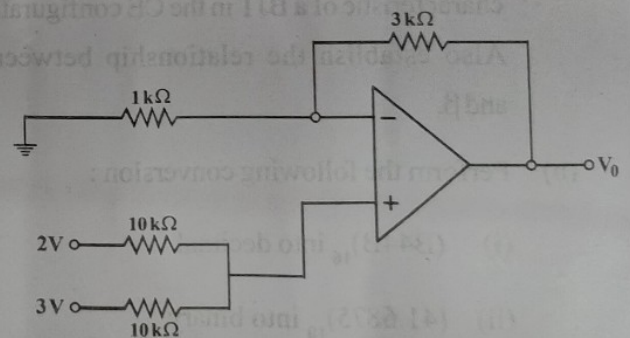
(iv) V_S

(v) V_G

(vi) V_D



(c) Find the output voltage V_0 :



[P.T.O.]

5. (a) What are the universal gates? Implement XOR & XNOR gate using NAND gate only.
- (b) Simplify the following Boolean expressions to a minimum number of literals :
- $AB + A\bar{B}$
 - $\overline{(A+B)} \overline{(\bar{A} + \bar{B})}$
 - $ABC + \bar{A}B + AB\bar{C}$
 - $\bar{A}\bar{C} + ABC + A\bar{C}$
- (c) Draw the subtractor using OP-amp and derive the expression for its output voltage.
6. (a) Draw and explain the input and output characteristic of a BJT in the CE configuration. Also establish the relationship between α and β .
- (b) Perform the following conversion :
- $(B44B)_{16}$ into decimal
 - $(41.6875)_{10}$ into binary

- $(2653.47)_8$ into binary
 - $(22.3125)_{10}$ into hexadecimal
- (c) Explain working and output waveform of full wave bridge rectifier in details.
